

Md Rahman

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EDUCATION

City College of New York, Grove School of Engineering

New York, NY

B.E. in Mechanical Engineering

Expected May 2029

- GPA: 4.00/4.00 | Dean's List | S Jay Levy Fellowship (2026–2027) | Celia Portnoy Memorial Scholarship
- Coursework: Engineering Mechanics I/Statics, Computer-Aided Drafting, Calculus III, University Physics I & II
- In Progress: Mechanics of Materials, Dynamics, Computer Methods in Engineering, Differential Equations

Stuyvesant High School — New York, NY — Advanced Regents Diploma, 2025

TECHNICAL SKILLS

CAD & Fabrication: SolidWorks, Onshape, Parametric Modeling, Engineering Drawings, 3D Printing, Machining

Analysis & Programming: MATLAB, Python, Excel, Arduino/Embedded C, Data Acquisition, PID Control, Git

Hardware & Testing: Design for Additive Manufacturing, Test Fixture Design, Sensor Integration, BOM

ENGINEERING EXPERIENCE

CarbonCLAIR — Environmental Technology Startup

New York, NY

Engineering Intern

June 2026 – Present

- Design a vacuum chamber for a 13.56 MHz capacitively coupled RF plasma reactor in argon at 30–150 mTorr, sizing chamber clearances and electrode gap from Debye length and verifying 8–25x buckling margins under 1 atm, documenting the design in a 31-page report and 23-figure engineering drawing set.
- Source a procurement BOM across 10+ vacuum suppliers, catching two feedthroughs marketed as RF-rated but specified at 450 kHz, not the required 13.56 MHz, and cost-model a \$19K staged build against a \$75K+ research-grade configuration.
- Build a droplet-detection subsystem for an electrohydrodynamic test rig: 3D-printed fixture, Arduino Uno, and laser-photodiode beam-break sensor with firmware that triggers a Photron high-speed camera at droplet entry, timed so its short buffer records only the droplet-plasma interaction.
- Assemble a 20+ component SolidWorks model of the 6 in. cube chamber, flanges, feedthroughs, and viewports, catching a fastener interference where 1/4-20 screws from adjacent plates collided inside a shared corner post, and respecifying hardware before ordering.

BioWearLab — Grove School of Engineering

New York, NY

Undergraduate Research Assistant, Mechanical Engineering

Jan 2026 – Present

- Engineer BCC lattice metamaterials in SolidWorks for 3D-printed TPU-graphene pressure sensors, optimizing unit-cell geometry across 8+ iterations to hit 0–50 N and 0–100 N gait-force ranges within 5% of target stiffness.
- Design and print test fixtures, compression mounts, and loading rigs on a Bambu Lab P1P (± 0.2 mm verified via calipers), enabling repeatable sensor characterization under compression, shear, and cyclic loading.
- Maintain Python and MATLAB pipelines analyzing 500+ data points per test cycle, generating force-displacement plots and summary tables that cut weekly reporting time roughly 40%.

FRC Team 694 (StuyPulse)

New York, NY

Mechanical Lead

Sept 2024 – June 2025

- Redesigned an overweight drivetrain and two-stage gearbox that flexed under competition loads, cutting 4.8 lbs (4%) from a 120 lb robot via plate pocketing and ribbed geometry in SolidWorks/Onshape, verified by hand calculation and bench load testing.
- Tuned PID gains across 15+ iterations on encoder and IMU feedback to reach ± 3 cm autonomous path error over a 5 m traverse, integrated with FDM-printed intake and manipulator assemblies.

LEADERSHIP & OPERATIONS

Khan's Tutorial | Branch Manager | Queens, NY

Jan 2026 – Present

- Oversee operations, enrollment, and instruction across 6 branches serving 180+ students and 8 instructors; closed 50+ enrollments totaling \$30,000+ in revenue and cut missed sessions 20% via attendance tracking.

MIST New York | Associate Chair | New York, NY

Sept 2025 – Present

- Direct day-of logistics for a 1,150-attendee tournament at Adelphi University, coordinating 80+ volunteers across 30+ competition events delivered 8% under budget.